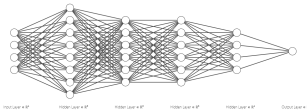
PAC-Bayesian Learning of Aggregated Binary Activated Neural Networks with Probabilities over Representations

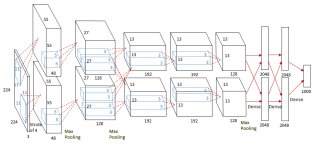
Benjamin Leblanc Louis Fortier-Dubois Gaël Letarte
Pascal Germain François Laviolette

June 6th, 2023

Introduction

- Neural networks: why so difficult to analyse?
- Simplification as a solution
- Probability distribution over the weights of the model
- Based on the work of *Letarte et al.*[1]

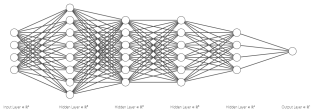


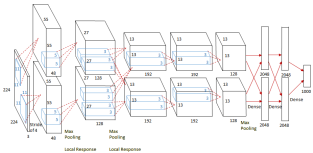


[1] Gaël Letarte, Pascal Germain, Benjamin Guedj, and François Laviolette. Dichotomize and generalize: PAC-Bayesian binary activated deep neural networks. NeurIPS, 2019.

Introduction

- PAC-Bayesian theory
  - Non-trivial generalization bound
- Contributions
  - Get rid of an approximation
  - Constant-time prediction relative to network depth





Plan of the presentation

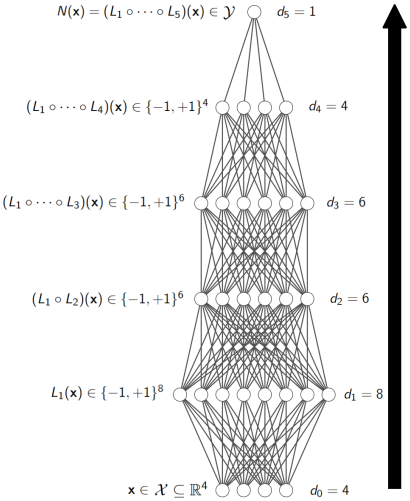
1. Preliminary notions
2. Neural networks aggregations
3. Experimentation
4. Question period



Preliminary notions

Neural networks

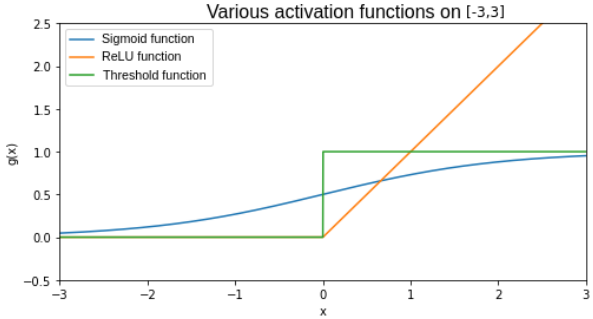
- Fully connected networks
- Binary classification tasks
- $F(\mathbf{x}) = (L_1 \circ \cdots \circ L_l)(\mathbf{x})$
- $L_k(\mathbf{x}) = g_k(\mathbf{W}_k\mathbf{x} + \mathbf{b}_k)\ \forall k \in \{1, \ldots, l\}$



Binary activation

Why?

- Computation time savings
- Gains in memory required
- Networks less prone to overfitting [1] [2]



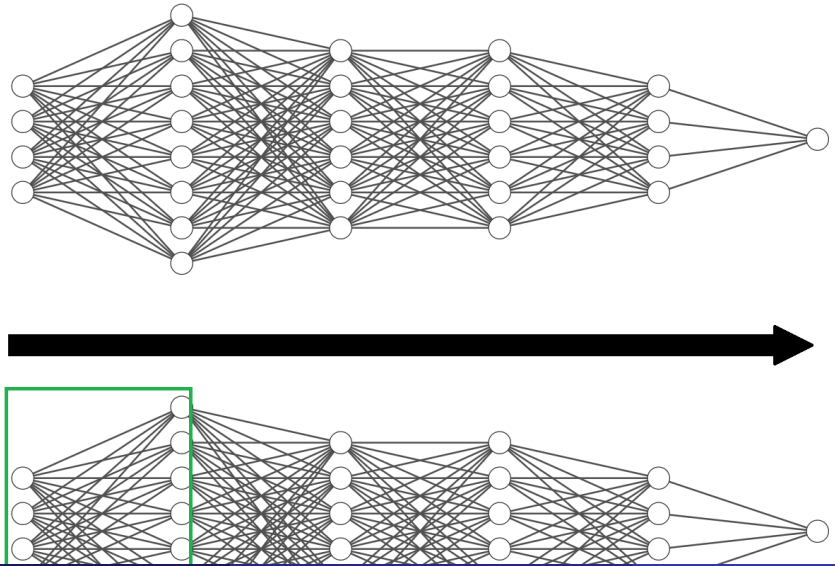
[1] J. Lin, C. Gan, and S. Han, "Defensive quantization: When efficiency meets robustness". CoRR, 2019
[2] M. Courbariaux, I. Hubara, D. Soudry, R. El-Yaniv, and Y. Bengio, "Binarized neural networks: Training deep neural networks with weights and activations constrained to +1 or -1". NIPS, 2016.

Dissecting binary activated neural networks

Definition : The parts of a binary activated networks

- The first part is L_1
- The second part is $(L_2 \circ \cdots \circ L_l)$

Binary activations - A visualization



Dissecting binary activated neural networks

Definition : The parts of a binary activated networks

- The first part (L_1) assigns a *group* to a given example
- The second part ($L_2 \circ \cdots \circ L_l$) assigns a *prediction* to each group

We have $L_1 : \mathcal{X} \to \{-1, +1\}^{d_1}$ and $(L_2 \circ \cdots \circ L_l) : \{-1, +1\}^{d_1} \to \mathcal{Y}$

Neural networks aggregation

Neural networks aggregation

Bayesian aspect

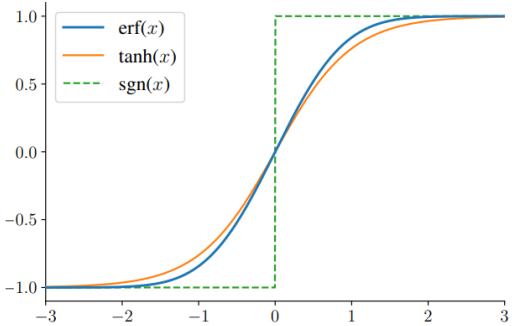
We suppose the weights $B = \langle \mathbf{W}_k \rangle_{k=1}^{l}$ to follow a Gaussian probability distribution, centred on B and with isotropic covariance matrix.

The output of a neuron

The expected output of a sign function applied Gaussian distributed variable:

$$\mathop{\mathbf{E}}_{\mathbf{v}\sim\mathcal{N}(\mathbf{w},\mathbf{I})} \operatorname{sgn}(\mathbf{v}\cdot\mathbf{a}) = \operatorname{erf}\left(\frac{\mathbf{w}\cdot\mathbf{a}}{\sqrt{2}||\mathbf{a}||}\right) ,$$

where $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$ is the gaussian error function.



Neural networks aggregation

Probability of observing a given output for a given neuron

The binary activation acts as a Bernoulli distribution. $P(s_i \in \{-1, +1\})$ is given by

$$\Pr(L_{k;i}(\mathbf{a}) = s) = \frac{1}{2} + \frac{s}{2}\mathrm{erf}\left(\frac{\mathbf{w}_{k,i} \cdot \mathbf{a}}{\sqrt{2}\|\mathbf{a}\|}\right), i \in \{1, \ldots, d_k\}, k \in \{1, \ldots, l\}.$$

Probability of observing a given output for a given layer

Let $\Pr((L_1 \circ \cdots \circ L_k)(\mathbf{a}) = \mathbf{s}) = L^{\mathbf{s}}_{1,k}(\mathbf{a})$. An iterative computation can be used:

$$\Pr(L^{\mathbf{s}}_{1,k}(\mathbf{a})) = \begin{cases} \prod_{i=1}^{d_1} \Pr(L^{s_i}_{1,i}(\mathbf{a})) & \text{si } k = 1, \\ \sum_{\bar{\mathbf{s}}} \Pr(L^{\mathbf{s}}_{1,k}(\mathbf{a}) \mid L^{\bar{\mathbf{s}}}_{1,k-1}(\mathbf{a})) \Pr(L^{\bar{\mathbf{s}}}_{1,k-1}(\mathbf{a})) & \text{sinon.} \end{cases}$$

The exact computation of ABNet

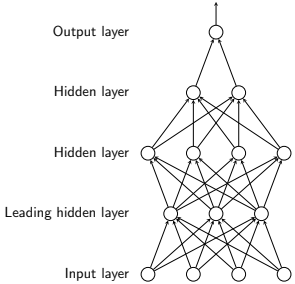
Dynamic programming approach

$$\text{First layer: } \mathbf{P}_1(\mathbf{x}) = \left[\Pr(L_1^{\mathbf{s}}(\mathbf{a})) \right]_{\mathbf{s} \in R_1}$$

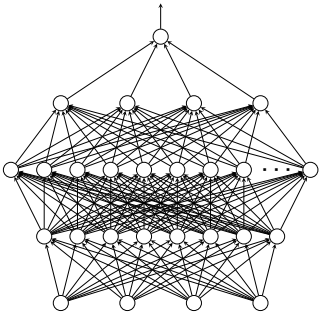
$$\begin{aligned} \text{Subsequent layers: } \mathbf{P}_k &= \left[\Pr(L_{1,k-1}^{\mathbf{s}}(\mathbf{a})) \right]_{\mathbf{s} \in R_k} \\ &= \mathbf{\Psi}_k \cdot \mathbf{P}_{k-1} \ , \end{aligned}$$

$$\text{with } \mathbf{\Psi}_k = \left[\Pr(L_{1,k}^{\mathbf{s}}(\mathbf{a}) \mid L_{1,k-1}^{\bar{\mathbf{s}}}(\mathbf{a})) \right]_{\mathbf{s} \in R_k, \bar{\mathbf{s}} \in R_{k-1}}$$

Neural networks aggregation



(a) A typical neural network



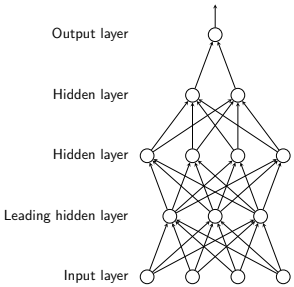
(b) The architecture of **ABNet**, with underlying network from (a).

Neural networks aggregation

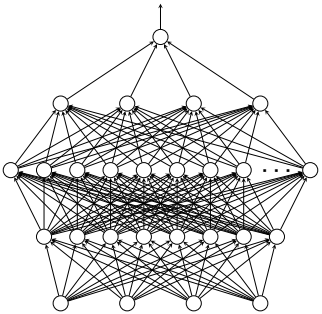
Compact ABNet

$$\begin{aligned}\Pr(L^s_{1,l}(\mathbf{a})) &= s \cdot (\mathbf{\Psi}_l(\mathbf{\Psi}_{l-1}(\dots \mathbf{\Psi}_3(\mathbf{\Psi}_2\mathbf{P}_1(\mathbf{x}))\dots))) \\ &= \underbrace{s \cdot \mathbf{\Psi}_l \cdot \mathbf{\Psi}_{l-1} \cdot \dots \cdot \mathbf{\Psi}_3 \cdot \mathbf{\Psi}_2}_{\mathbf{H}_s} \cdot \mathbf{P}_1(\mathbf{x}))\end{aligned}$$

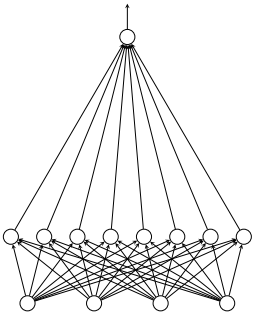
Neural networks aggregation



(a) A typical neural network.



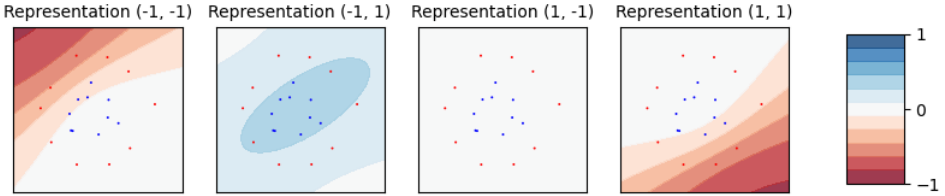
(b) The architecture of **ABNet**, with underlying architecture from (a).



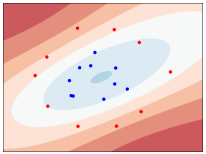
(c) The structure of **compact ABNet**, obtain from the network presented in (b).

Experimentations

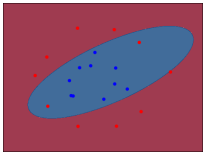
ABNet - a pratical case



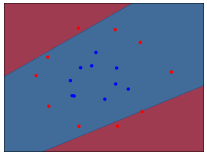
(a) For different values of $\mathbf{x}$, the probability of obtaining a certain representation as an output of the first hidden layer, obtained with $\mathbf{P}_1(\mathbf{x})$, multiplied by $\mathbf{H}$.



(b) The output of the aggregation $\Pr(L_{1,l})$.



(c) ABNet as a classifier.



(d) Unique network, centered on the *a posteriori* means of the Gaussian probability distribution.

Experimentations

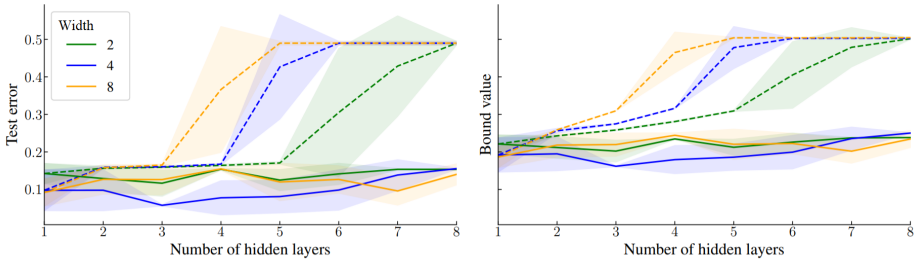


Figure: Impact of the depth for **PBGNet (dotted)** and **ABNet (continuous)** on the test error and the bound value depending on the width of the network on mnistLH. 5 random seeds.

Experimentations

Dataset	Model	$L-1$	d	Bound	Error_S	Error_T
ads	PBGNet	3	2	0.192 ± 0.004	0.140 ± 0.004	0.141 ± 0.012
	ABNet	3	2	0.192 ± 0.004	0.140 ± 0.004	0.141 ± 0.012
	PBGNet_ℓ	3	4	1.000 ± 0.001	0.018 ± 0.005	0.026 ± 0.004
	ABNet_ℓ	3	4	0.887 ± 0.064	0.015 ± 0.003	0.026 ± 0.003
	EBP	2	2	–	0.003 ± 0.002	0.040 ± 0.008
	BC	1	4	–	0.025 ± 0.005	0.031 ± 0.004
	BNN	1	8	–	0.037 ± 0.002	0.038 ± 0.004
mnistLH	PBGNet	1	8	0.186 ± 0.028	0.091 ± 0.037	0.092 ± 0.036
	ABNet	3	4	0.162 ± 0.001	0.056 ± 0.001	0.058 ± 0.002
	PBGNet_ℓ	3	8	1.000 ± 0.000	0.018 ± 0.003	0.038 ± 0.002
	ABNet_ℓ	2	8	0.998 ± 0.003	0.025 ± 0.008	0.042 ± 0.006
	EBP	3	8	–	0.016 ± 0.002	0.043 ± 0.002
	BC	2	8	–	0.023 ± 0.002	0.035 ± 0.001
	BNN	1	2	–	0.123 ± 0.005	0.133 ± 0.004

Questions / comments

Thank you for listening :) !